

Yerrapati Venkata Subbaiah

Email: venkatasubbaiahvs2003@gmail.com

Location: Surat, Gujarat, India

EDUCATION

Sardar Vallabhbhai National Institute of Technology

CGPA : 8.59/10.0

December 2020 – June 2025

Surat, Gujarat, India

Integrated Master of Science in Mathematics

Sri Vagdevi Junior College

Marks : 974/1000

June 2018– February 2020

Kodad, Telangana, India

Board of Intermediate Education, Telangana

Teja Vidyalaya

CGPA : 9.3/10.0

March 2018

Kodad, Telangana, India

Board of Secondary Education, Telangana

RESEARCH EXPERIENCE

Summer Research Intern | PI: Prof. Manoj Kummini

Chennai Mathematical Institute

June 17, 2024 - July 25, 2024

Chennai, Tamil Nadu, India

- I worked on understanding Symmetric Groups Representations.
- Specifically on *relation between representations and properties of automorphisms of Symmetric Groups*.

Summer Research Intern | PI: Dr. Amiya Kumar Mondal

Indian Institute of Science Education and Research

May 01, 2024 - June 15, 2024

Berhampur, Odisha, India

- A reading project on local fields where I got introduced to p -adic numbers and their properties.
- I got introduced to Ramification groups which plays a significant role in representations.

Winter Research Intern | PI: Prof. Stefan Keppeler

Universität Tübingen

December 2023 - February 2024

Virtual – Tübingen, Germany

- This is a self reading project on Representation Theory of Finite Groups.
- This internship really excited me. As when I got to know, understanding an abstract structure using matrices.

Summer Internship Program | PI: Dr. Kumar Balasubramanian

Indian Institute of Science Education and Research

May 2023 – July 2023

Bhopal, Madhya Pradesh, India

- I worked on Matrix Lie Groups with the understanding of Group Theory and Linear Algebra.
- I just got Introduced to Lie Algebras and worked on examples like $\mathfrak{gl}(n, \mathbb{F})$, $\mathfrak{sl}(n, \mathbb{F})$.

Winter Research Intern | PI: Dr. Ankur Kanaujiya

National Institute of Technology, Rourkela

December 2022

Rourkela, Odisha., India

- I got introduced to portfolio management and risk prediction models like GARCH, IGARCH etc.
- I have also worked on optimizing the portfolio with the help of Sharpe Ratio etc and analysing the portfolio with the help of VaR.

PUBLICATIONS

Research Interest : Representation Theory

1. Yerrapati, V. S., Dixit, R., & Shukla, A. K. (2026). *Twisted Frobenius-Schur Indicators and Character Degree Sums in Dihedral Groups*. arXiv preprint arXiv:2605.22127.
2. *On the Sum of degrees of all Irreducible Representations and Involutions of an Automorphism*. (on-going)

TRAINING PROGRAMMES

Summer Workout in Mathematics

Tata Institute of Fundamental Research - CAM

June 01, 2022 - June 28, 2022

Benguluru, Karnataka., India

- It is a training programme on Real Analysis, Linear Algebra, Statistics and Python Programming.
- I have worked on COVID Simulation Project and started learning Real Analysis, Linear Algebra where I left in MTTS Camps.

Initiation into Mathematics, MTTS

Bihar Camp

March 05 - 20, 2022

Virtual

- A training programme on Foundations of Mathematics, Linear Algebra, Real Analysis. organized by MTTS Trust, funded by National Board of Higher Mathematics (NBHM).
- This programme helped me to improve my communication skills to present my views on a problem.

Follow-up Course on Linear Algebra

OFCM - Follow up, MTTS

February 2022

Virtual - India

- It is a training programme on Linear Algebra organized by MTTS Trust and funded by National Board of Higher Mathematics (NBHM)
- I got introduced to Linear Algebra. After this I have learnt linear algebra just reading and through imagination.

Follow-up Course on Real Analysis

OFCM - Follow up, MTTS

January 2022

Virtual - India

- It is a training programme on Real Analysis organized by MTTS Trust and funded by National Board of Higher Mathematics (NBHM)
- I got introduced to Real Analysis in which I learnt how to approach a problem

Online Foundational Course in Mathematics

OFCM, MTTS

September 2021 - October 2021

Virtual - India

- It is a training programme on Foundations of Mathematics organized by MTTS Trust and funded by National Board of Higher Mathematics (NBHM)
- I have understood how to work with Statements, Sets, Functions.

PROJECTS

TeX Newsletter and Magazine Automation using Python

Department Project

- Get's the data from user and generates tex files according to the standard newsletter and magazine of Department.
- Along with Rahul Maurya, Rajarapu Mahesh.

National Mathematics Day Web Page

Publicity Project | NMD 2022

- A web page where one can find the events organized on national mathematics day 2022.
- Along with Abhijeet Bansood, Rahul Maurya

Newsletter using LaTeX

Exploratory Project | Newsletter Committee

- Newsletter of the Department to show case the ongoing of the department.

Covid Simulation

Exploratory Project | P.I: Dr Venkatershwaran Krishnan

- Spread of Covid 19. When a person affected in a community.
- Simulated the situation where people wear masks and do not wear masks.

Student Chapter Portal

Exploratory Project | **on-going**

- A Portal where student chapters of SVNIT use to get permissions, budget sanctions etc. from the administration.

TECHNICAL SKILLS

Programming Skills:

- Proficient in SageMath, C, Python, and Matlab for algorithm development and data analysis.

Version Control Tools:

- Skilled in using Git and GitHub for collaborative coding and efficient version control.

Web Development:

- Familiar with web development technologies including HTML, CSS, and JavaScript, enabling interactive web-based projects.

Other Skills:

- Proficient in typesetting documents using LaTeX for creating professional reports and documents.
- Good Knowledge of Designing and working with software like Adobe Indesign, Adobe Illustrator, Canva.

SERVICES

- *Mentor* at Online Foundation Course in Mathematics 2025, organized by MTTS Trust, funded by National Board of Higher Mathematics (NBHM).
- *Mentor* at Online Foundation Course in Mathematics 2024, organized by MTTS Trust, funded by National Board of Higher Mathematics (NBHM).
- *Volunteer* of Indian Space Research Organization (ISRO) Exhibition held at SVNIT, Surat.
- *Student organizing member* of the following Expert Talks held at SVNIT, Surat.
 - Prof. V. P. Saxena, Former Vice-chancellor of Jiwaji University, Gwalior.
 - Prof. K. N. S. Viswanadham, Dean Academic and Professor of Mathematics, NIT Warangal.
- *Organising member* of Curry Leaf Days 2022, Curry Leaf, Math Club of MTTS Alumni.
- *Organising member* of Article Writing Workshop on 24/08/2022, DoMH, SVNIT, Surat
- *Student organizing member* of Valedictory Session, SWIM programme 2022, TIFR-CAM, Bangalore.
- *Organising member* of National Mathematics Day 2021 Celebrations, DoMH, SVNIT Surat.

RESPONSIBILITIES

- **Member**, Initiative Committee, DoM, SVNIT Surat during Academic Year 2023-24.
- **Head**, Research Team and Think Tank, SVNIT Surat during Academic Year 2022-23.
 - Gist: I have worked on some of the academic policies such as Nptel Credit Transfer, Introducing Diamond and Textile Engineering Courses, Teaching Methodology.
- **Chief Editor**, AMaThing Committee, DoMH SVNIT Surat during January 2023 - May 2023.
 - Gist: Magazine in which is collection of articles meeting the standards of UGC.
- **Chief Editor**, Pramiti Committee, DoMH SVNIT Surat during May 2022 - May 2023.
 - Gist: An annual report of Department showcasing the achievements of faculty, students achieved during the academic year.
- **Chief Editor**, Newsletter Committee, DoMH SVNIT Surat during March 2021 - May 2023.
- **Co-Convener**, National Mathematics Day 2022, DoMH, SVNIT Surat.

INITIATIVES

- **Departmental Orientation for freshmen (J)**
 - Event would provide a glimpse what it means to study in Dept. and activities conducting by Dept.
- **Department Newsletter (J)**
 - Triannual document show casing ongoing of the Dept.
- **Department Handbook (J) and Brochure (J)**
 - Document show casing information mandatory for freshmen.
- **Pramiti (J)**
 - Document provided to the public consisting of annual activities.
- **Training Programme and Group Discussions (I)**
 - The fellow mates who are interested in doing maths, come together to discuss and as well as training the juniors.

Note: J-Joint work mostly with Sai Charan Gannamaneni, Theophilus Gera, Mahesh Rajarapu, Ekta Jain etc.

ACHIEVEMENTS/AWARDS/HONORS

- Cleared GATE 2026 (AIR-938) in Mathematics.
- Cleared GATE 2024 (AIR - 1150) in Mathematics.
- JEE Mains 2020, Secured Rank in the merit List: Top 48,000 among 1 Million candidates.
- One of the people among the top 15 contests in the Sakshi Math Bee Competition in the year 2018.
- Received Certificate of Merit in Green Olympiad in the year 2016.
- Received Consolation prize in sculpting in Balotsav 2016 (National Level) at Kothagudem and first prize in sculpting in Balotsav (State Level) at Munagala in the years 2016 and 2017.

EXTERNAL COURSES/READING PROJECTS

- Representation Theory of Finite Groups
- Representation Theory of Symmetric Groups
- Introduction to Lie Algebras

LANGUAGES

- Telugu (Bilingual Proficiency)
- English (Working Profeciency)
- Hindi (Conversational)

REFERENCES

- **Prof. Manoj Kummini**
Professor, Department of Mathematics
Chennai Mathematical Insitute, Chennai, Tamil Nadu, India
E-Mail Id: mkummini@cmi.ac.in
- **Dr. Venkateshwaran Krishnan**
Associate Professor, Centre for Applicable Mathematics
Tata Institute of Fundamental Research, Bengaluru, Karnataka, India
E-Mail Id: vkrishnan@tifrbang.res.in

- **Prof. Ajay Kumar Shukla**

Professor, Department of Mathematics, SVNIT, Surat, Gujarat

E-Mail Id: aks@amhd.svnit.ac.in

- **Dr. Jayesh M. Dhodiya**

Associate Professor and Head, Department of Mathematics, SVNIT, Surat, Gujarat

E-Mail Id: jmd@amhd.svnit.ac.in

- **Dr. Shivam Bajpeyi**

Assistant Professor, Department of Mathematics, SVNIT, Surat, Gujarat

E-Mail Id: shivambajpeyi@amhd.svnit.ac.in

TALKS/CONFERENCE/WORKSHOP ATTENDED

1. Participated in OneMath World School 2026 Topological Data Analysis and Applications at Chennai Mathematical Institute, Chennai during February 20-28, 2026.
2. Attended International Conference on Recent Advances in Applied Mathematics and Computing at MIT, Bengaluru during January 7-9, 2026.
3. Attended International Conference on Algebra and Number Theory: 100th Birth Anniversary of Professor R. P. Bambah (ICANT-2025) at Panjab University, Chandigarh during September 30 - October 01, 2025.
4. Attended Ramanujan Mathematical Society Conference at Christ University, Bengaluru during December 27-29, 2024.
5. Participated in AIS on Representation Theory of Quivers and BGG Category O for semi simple Lie Algebras, IIT Kanpur, Online Mode during December 2-21, 2024.
6. Participated in National Workshop on Algebra and Analysis, DoM, NIT Tiruchy during December 16-20, 2024.
7. Attended Algebra Day Event, IIT Madras on June 25, 2024.
8. Attended INDIAN WOMEN AND MATHEMATICS (IWM), Annual Conference at IISER Bhopal during July 13-15, 2023. Funded by National Board for Higher Mathematics, DAE, Govt. of India and Committee for Women in Mathematics, IMU
9. Participated in seminar on **Autonomous Air Vehicles-Technologies & Applications**, an annual event jointly conducted by **Design Division of Aeronautical Society of India** and **Aeronautical Development Establishment (DRDO)**, Bangalore during January 6 - 7, 2023.
10. Participated in **The 14th Medieval History Journal Annual Lecture 2022** on **Reason vs Religion in Medieval India: Mainly Based on Evidence in Persian** on 19th November 2022.
11. Attended webinar on **Analysis & Differential Equations** organised by Department of Mathematics in collaboration with Department of Statistics, Panit Deendayal Upadhyaya Adarsha Mahavidyalaya (PDUAM) Eralgool, Karimganj, Assam during March 1-2 2022.
12. Participated in “2nd International Conference on Mathematical Modelling and Simulation in Physical Sciences” organized by Department of Mathematics and Humanities, SVNIT Surat during 5-6 February, 2022.
13. Attended webinar on **National Mathematics Day In the memory of Srinivasa Ramanujan** Organised by Department of Mathematics, Amity School of Applied Sciences (ASAS) on 22nd December 2021.